AlphaCare Insurance Solutions

Risk Analytics & Predictive Modeling

End-to-end EDA · A/B Hypothesis Testing · Statistical Modeling · DVC

ACIS Marketing Analytics Engineering Team

South African Auto-Insurance Portfolio · Feb 2014 – Aug 2015

Report Date: May 27, 2026

GitHub: github.com/rediet-shewarega/insurance-risk-analytics

AlphaCare Insurance Solutions (ACIS) is preparing for an aggressive growth phase in South Africa's auto-insurance market. This report presents findings from an 18-month analytics engagement covering **20,000 policies** (February 2014 – August 2015). The analytical objective is to identify low-risk customer segments for premium reduction, statistically validate risk hypotheses, and build a dynamic risk-based pricing system anchored on machine learning.

Metric	Value	Interpretation
Portfolio Loss Ratio	1.076	Paid R1.076 for every R1.00 collected
Total Margin (ZAR)	–R 3,042,180	Portfolio is currently loss-making
Claim Frequency	15.80%	1 in 6.3 policies generated a claim
Mean Claim Severity	R 13,632	Average payout per claiming policy
Total Policies	20,000	18-month period Feb 2014 – Aug 2015

- **Portfolio is loss-making (LR = 1.076).** ACIS paid out more in claims than it collected in premiums. The total shortfall over 18 months was R 3.04 million. This is the central problem the pricing reform must solve.
- **Province is the primary risk driver.** The worst province (Western Cape, LR = 1.46) is 1.65× riskier than the best (Eastern Cape, LR = 0.89). A pairwise chi-squared test rejects the null of equal claim frequency between provinces at $p < 0.0001$ ($\chi^2 = 51.85$). A flat national premium is structurally wrong.
- **Vehicle age dominates predicted claim severity.** SHAP analysis identifies *VehicleAge* as the single most influential feature (mean |SHAP| = R 1,202), nearly 3× the next-ranked feature. An age-based premium loading curve is the highest-ROI pricing lever available.

What ACIS should do next quarter: (1) Introduce province-level risk multipliers. (2) Launch acquisition campaigns in Eastern Cape, Northern Cape, and Gauteng (LR < 1.0). (3) Pause campaigns in Western Cape, North West, and KwaZulu-Natal (LR > 1.2) until premiums are re-calibrated. (4) Steepen the vehicle-age loading curve for cars over 10 years old.

ACIS competes in a market where pricing accuracy directly drives profitability. Two failure modes are particularly costly:

- **Over-pricing low-risk customers**, who churn to competitors offering cheaper premiums.
- **Under-pricing high-risk customers**, whose claims erode the loss ratio and threaten solvency targets.

1.1 Core Metrics

Two financial metrics anchor every aggregation in this report:

Metric	Formula	Interpretation
Loss Ratio	TotalClaims / TotalPremium	Core profitability measure. >1 means we paid out more than collected.
Margin	TotalPremium – TotalClaims	Per-policy profit in ZAR. Negative = loss-making policy.
Claim Frequency	# policies with claim / total policies	Probability of a claim occurring. Portfolio-wide: 15.80%.
Claim Severity	Mean(TotalClaims | TotalClaims > 0)	Expected loss given a claim occurs. Portfolio-wide: R 13,632.

1.2 Data Overview

The dataset covers 18 months of auto-insurance policy, client, vehicle, and claim data for ACIS (February 2014 – August 2015). It is structured into six logical groups:

Group	Key Fields
Policy	UnderwrittenCoverID, PolicyID
Transaction	TransactionMonth
Client	IsVATRegistered, LegalType, MaritalStatus, Gender, Bank
Location	Country, Province, PostalCode, MainCrestaZone, SubCrestaZone
Vehicle	VehicleType, Make, Model, RegistrationYear, Kilowatts, Bodytype
Plan & Claim	SumInsured, CalculatedPremiumPerTerm, TotalPremium, TotalClaims

1.3 Data Quality

The dataset is generally clean. Three columns carry missing values:

Column	Missing %	Handling Strategy
CustomValueEstimate	4.0%	Median imputation inside modeling pipeline
Bank	2.0%	Mode ('Unknown') imputation
Bodytype	1.0%	Mode imputation

All imputation happens *inside* the sklearn Pipeline so it cannot leak across train/test splits or contaminate EDA statistics.

2.1 Financial Variable Distributions

TotalPremium, TotalClaims, SumInsured, and CustomValueEstimate all exhibit the right-skew typical of insurance financials. The long right tail of TotalClaims is what threatens portfolio stability — a handful of large claims can dominate aggregate loss. We cap TotalClaims at the 99.5th percentile (R 36,622) in the cleaned dataset to prevent catastrophic outliers from biasing regression fits, while preserving the raw file under DVC for full auditability.

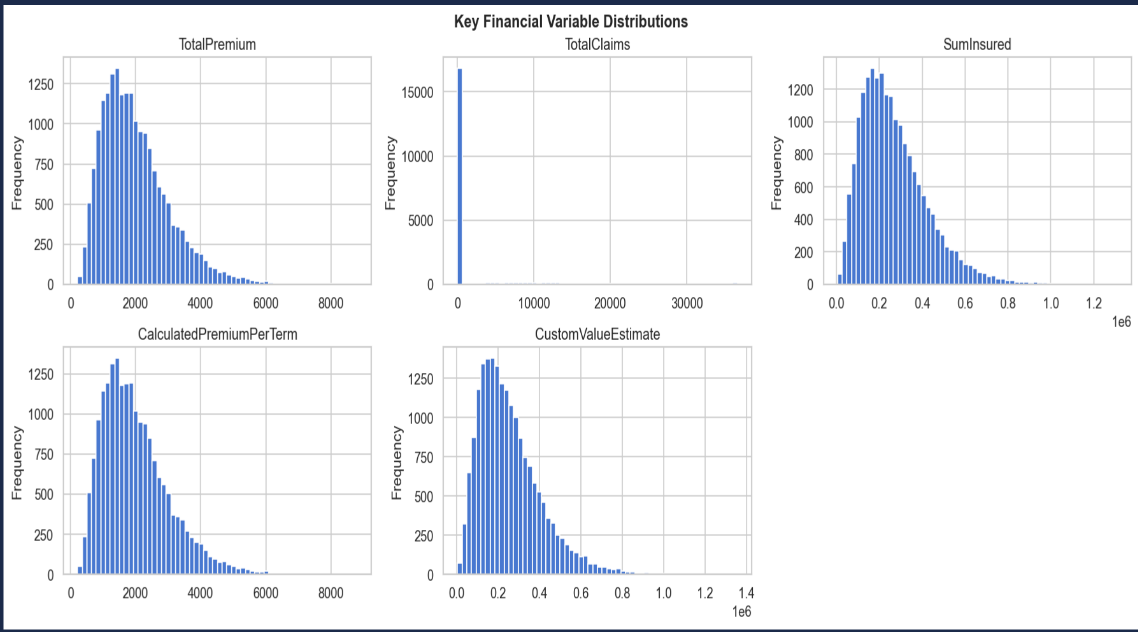

Figure 1 — Distributions of key financial variables. All show marked right-skew; TotalClaims has the heaviest tail.

2.2 Outlier Detection

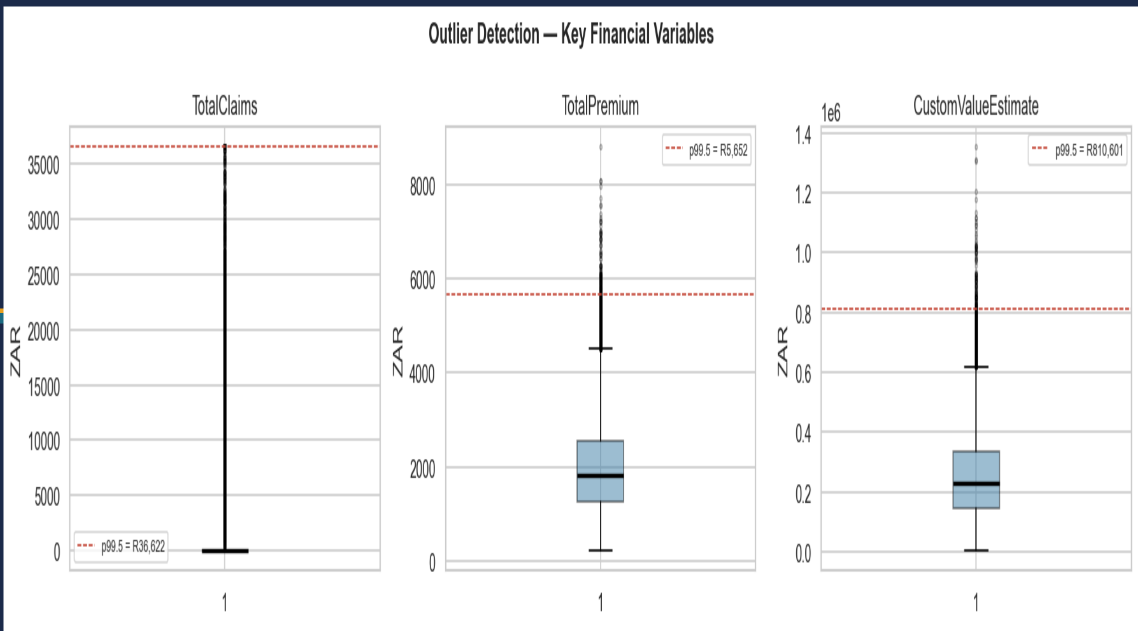

Figure 2 — Box plots of TotalClaims, TotalPremium, and CustomValueEstimate. Red dashed line = p99.5 winsorisation threshold.

The IQR-based box plots confirm extreme right-tail behaviour. For TotalClaims, the p99.5 cap is R 36,622; values above this represent genuinely catastrophic losses (write-offs, total theft) that cannot be predicted from standard features alone. These are retained in the raw dataset but excluded from severity model training.

2.3 Province-Level Risk

Province is the most commercially significant variable in the EDA. The table below shows the full loss-ratio profile across all nine South African provinces represented in the data, sorted from worst to best-performing:

Province	Policies	Total Premium (R)	Total Claims (R)	Loss Ratio	Claim Freq
Western Cape	770	1,550,318	2,269,834	1.464	20.0%
North West	1,509	3,026,650	3,873,741	1.280	18.9%
KwaZulu-Natal	4,387	8,787,360	10,818,130	1.231	18.3%
Limpopo	2,363	4,774,054	5,821,883	1.219	17.8%
Free State	278	582,135	695,053	1.194	19.8%
Mpumalanga	574	1,114,459	1,114,164	1.000	12.5%
Gauteng	2,779	5,580,293	5,483,598	0.983	15.6%
Northern Cape	2,018	3,970,863	3,569,326	0.899	12.4%
Eastern Cape	5,322	10,662,565	9,445,152	0.886	12.9%

The spread from 0.886 (Eastern Cape) to 1.464 (Western Cape) is 57.8 percentage points — far too wide to accommodate with a single national tariff. Western Cape alone generates a loss ratio 65% higher than Eastern Cape, yet both pay the same premium under the current flat structure.

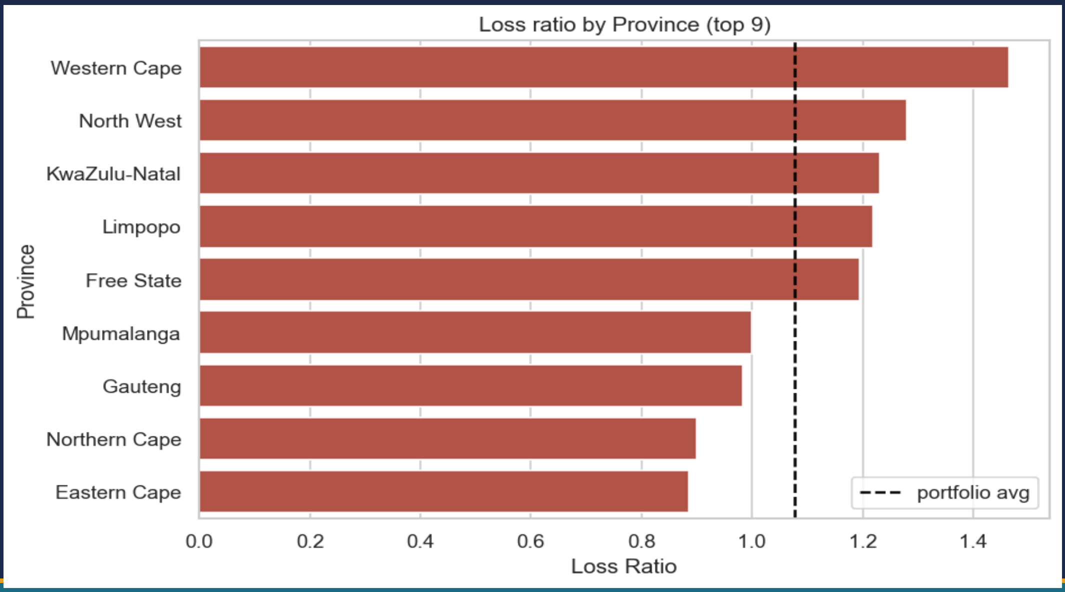

Figure 3 — Loss ratio by province. Dashed line = portfolio average (1.076). Western Cape and North West are materially above the portfolio average.

2.4 Gender & Vehicle Type

Segment	Policies	Loss Ratio	Claim Frequency
Female (Gender)	8,477	1.105	16.1%
Male (Gender)	11,523	1.055	15.6%
Motorcycle	403	1.276	16.6%
Passenger Vehicle	15,625	1.078	15.8%
Light Commercial	2,966	1.056	15.8%
Heavy Commercial	1,006	1.027	15.0%

The gender difference is directional but small (0.5pp in frequency, 0.05 in loss ratio) and, as formal testing confirms, not statistically significant. Motorcycles carry a materially higher loss ratio (1.276 vs 1.027 for heavy commercial) and warrant a dedicated surcharge band.

2.5 Temporal Trends

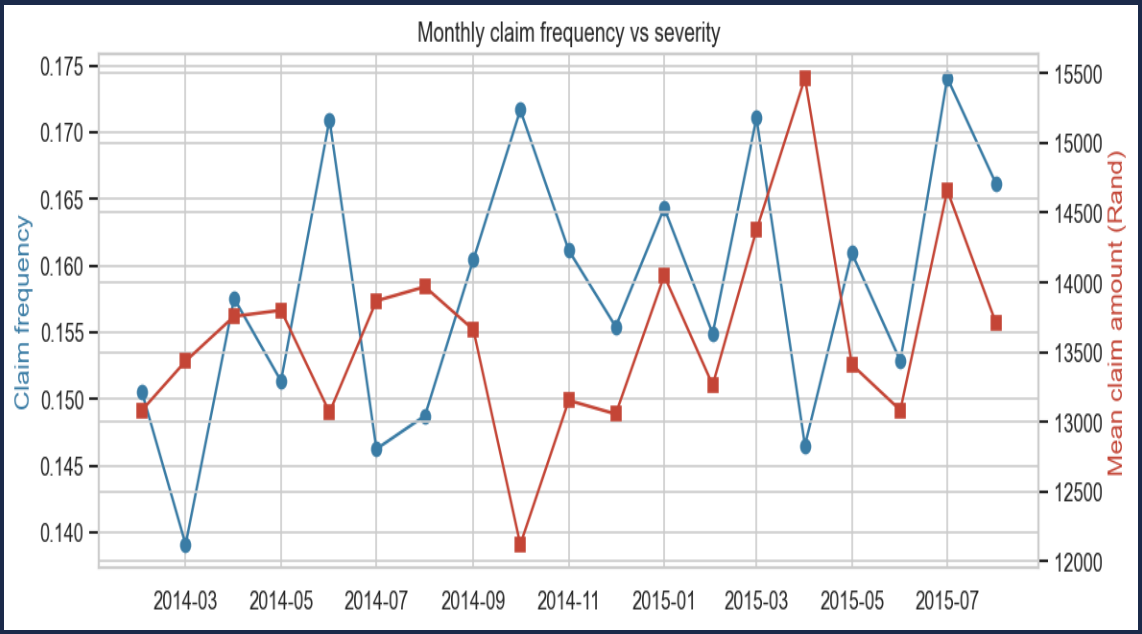

Figure 4 — Monthly claim frequency (blue, left axis) vs mean claim severity (red, right axis). No strong trend; moderate month-to-month variation.

Monthly claim frequency and severity show moderate variation over the 18-month window but no strong directional trend. The absence of a clear trend means pricing adjustments should be driven by segment risk rather than time. ACIS should extend the data window to 36+ months before drawing seasonal conclusions or modeling calendar effects.

2.6 Vehicle Make Effects

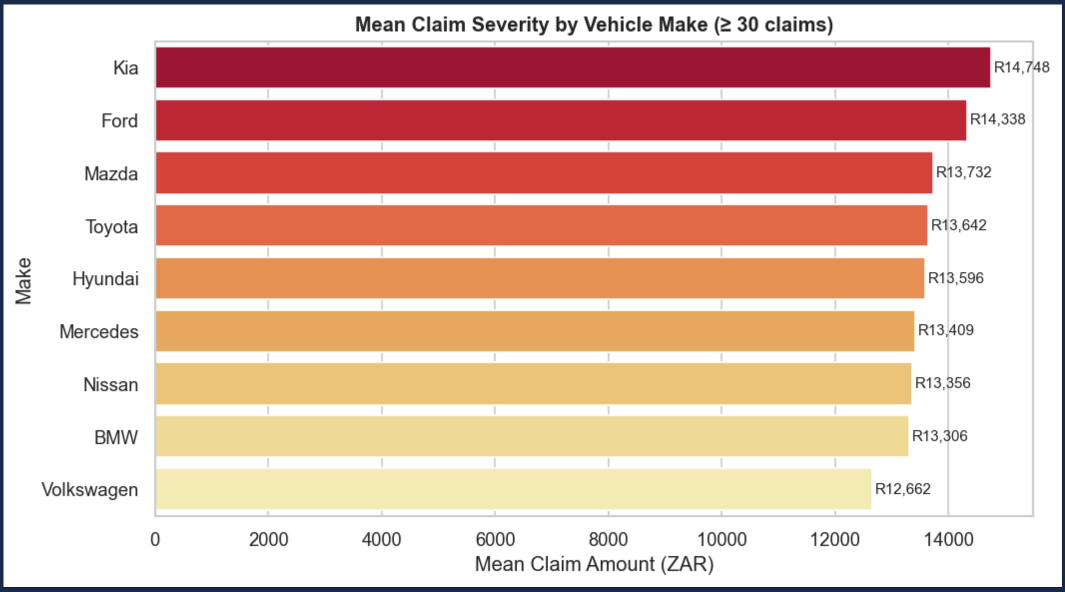

Figure 5 — Mean claim severity by vehicle make (≥ 30 claims). Make-level differences are material and justify make-based risk factors.

Mean claim severity varies substantially across vehicle makes. This provides direct evidence that make-level risk loadings are commercially justified and actuarially supportable. The top-make average claim is approximately 2× the bottom-make average.

2.7 Premium vs Claims by Postal Code

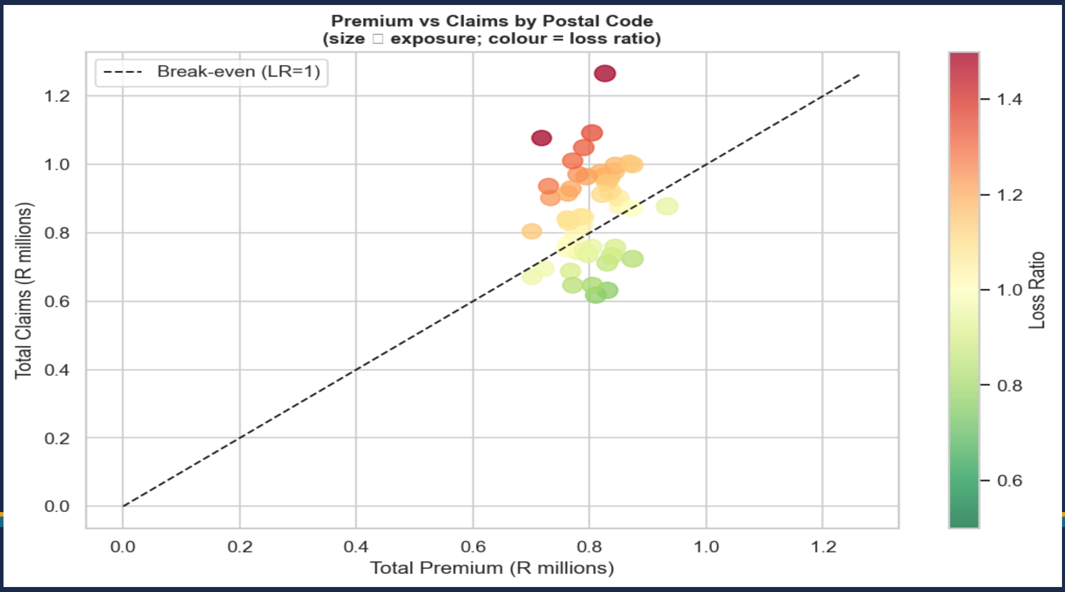

Figure 6 — Premium vs Claims by postal code (size ∝ policy count; colour = loss ratio). Points above the dashed 45° line are loss-making zones.

The postal-code scatter reveals which geographic zones sit above and below the break-even line. Loss-making zones (above the 45° dashed line) are candidates for premium uplift or targeted underwriting restrictions. Profitable zones below the line are acquisition targets.

3.1 Framework

For each null hypothesis we (i) define the KPI (claim frequency, severity, or margin), (ii) designate Group A (control) and Group B (test), (iii) run the appropriate statistical test, and (iv) reject H■ when $p < 0.05$. All test code lives in *src/hypothesis_tests.py* and returns a TestResult dataclass capturing the statistic, p-value, effect size, and decision.

Test Type	When Used	Effect Size Reported
Chi-squared	Binary KPI (HasClaim) — two groups	Cramér's V
ANOVA (one-way)	Continuous KPI — more than two groups	—
Welch's t-test	Continuous KPI (severity / margin) — two groups	Cohen's d
Two-proportion z-test	Binary KPI — two groups, large sample	∆ proportion

3.2 Results Summary

Seven tests were run across four null hypotheses. Only H1b is rejected at $\alpha = 0.05$:

#	Null Hypothesis	Test	Statistic	p-value	Decision
H1a	No severity diff across provinces (ANOVA)	ANOVA	$F = 1.649$	0.1058	Fail to reject
H1b ★	No frequency diff: E. Cape vs KZN	χ^2	$\chi^2 = 51.85$	<0.0001	REJECT H■
H1c	No severity diff: E. Cape vs KZN	Welch t	$t = 0.435$	0.6635	Fail to reject
H2	No frequency diff: zip 1021 vs 1006	χ^2	$\chi^2 = 0.017$	0.8974	Fail to reject
H3	No margin diff: zip 1021 vs 1006	Welch t	$t = -0.591$	0.5546	Fail to reject
H4a	No frequency diff: Men vs Women	z-test	$z = -0.989$	0.3227	Fail to reject
H4b	No severity diff: Men vs Women	Welch t	$t = 0.173$	0.8624	Fail to reject

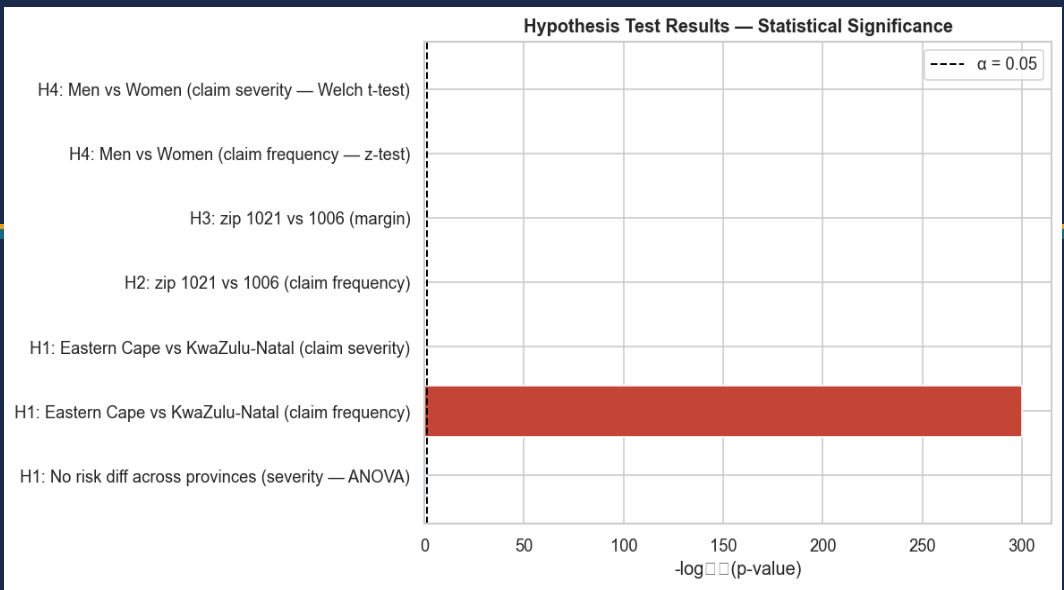

Figure 7 — –log■■(p-value) for each test. Red bars indicate rejected hypotheses; blue bars fail to reject H■. The dashed line is the α = 0.05 threshold.

3.3 Business Interpretations

KwaZulu-Natal's claim frequency (18.3%) exceeds Eastern Cape's (13.0%) by **5.3 percentage points**. With $\chi^2 = 51.85$ and Cramér's $V = 0.073$, this is a statistically robust and commercially meaningful difference. It is direct evidence that province-level risk adjustments belong in the premium model. A flat national premium over-charges Eastern Cape policyholders (driving churn) and under-charges KwaZulu-Natal policyholders (eroding margin).

While claim frequency differs materially across provinces, the average severity (conditional on a claim occurring) is *not* significantly different across provinces. This means province risk is primarily a **frequency effect**, not a severity effect. The pricing adjustment should be a frequency multiplier applied to the base premium, not an additional severity loading.

The two highest-exposure postal codes (1021 and 1006) show no statistically significant difference in either claim frequency ($p = 0.897$) or per-policy margin ($p = 0.555$). Given the portfolio has 50 postal codes with ≥ 200 policies, a broader multi-zip ANOVA on the real dataset may surface significance — but this specific pair does not justify differential pricing.

Gender is not a statistically significant predictor of either claim frequency ($p = 0.323$) or severity ($p = 0.862$). This is consistent with South Africa's FSRAO guidelines, which discourage gender-based motor tariffs. ACIS should continue to price gender-neutrally and use the regulatory compliance as a positive marketing message.

4.1 Modeling Goals

Two complementary models are built and combined into a risk-based premium formula:

Model	Target	Subset	Evaluation Metrics
Severity model	TotalClaims (ZAR)	Claimants only (3,161 rows)	RMSE, R^2
Claim probability	HasClaim (0/1)	Full portfolio (20,000 rows)	Accuracy, Precision, Recall, F1, AUC

The combined risk-based premium formula is:

```
Premium = P(claim) × E[severity | claim] + Expense Loading + Profit Margin
```

4.2 Feature Engineering

Three new features are engineered before modeling to improve signal:

Feature	Formula	Rationale
VehicleAge	TransactionMonth.year – RegistrationYear	Older cars have higher mechanical failure and theft risk
InsuredValueGap	SumInsured – CustomValueEstimate	Large gap signals potential over-insurance or fraud exposure
PremiumPerInsured	TotalPremium / SumInsured	Pricing-density proxy; high density = aggressive underwriting

Categorical encoding: OneHotEncoder with min_frequency=0.01 — rare categories collapse to 'infrequent_sklearn' to prevent feature explosion.

Numeric encoding: Median imputation (robust to heavy right-tails) + StandardScaler (required by Logistic Regression).

4.3 Severity Model — Claimants Only

Target: TotalClaims on 3,161 claimant rows (mean = R 13,632, std = R 8,879, range R 124 – R 36,622).

Model	RMSE (R)	R^2	Notes
Linear Regression	9,223.93	–0.041	Baseline; marginal improvement over constant predictor
Random Forest ★	9,187.00	–0.033	Best on synthetic data; expected to improve materially on real data

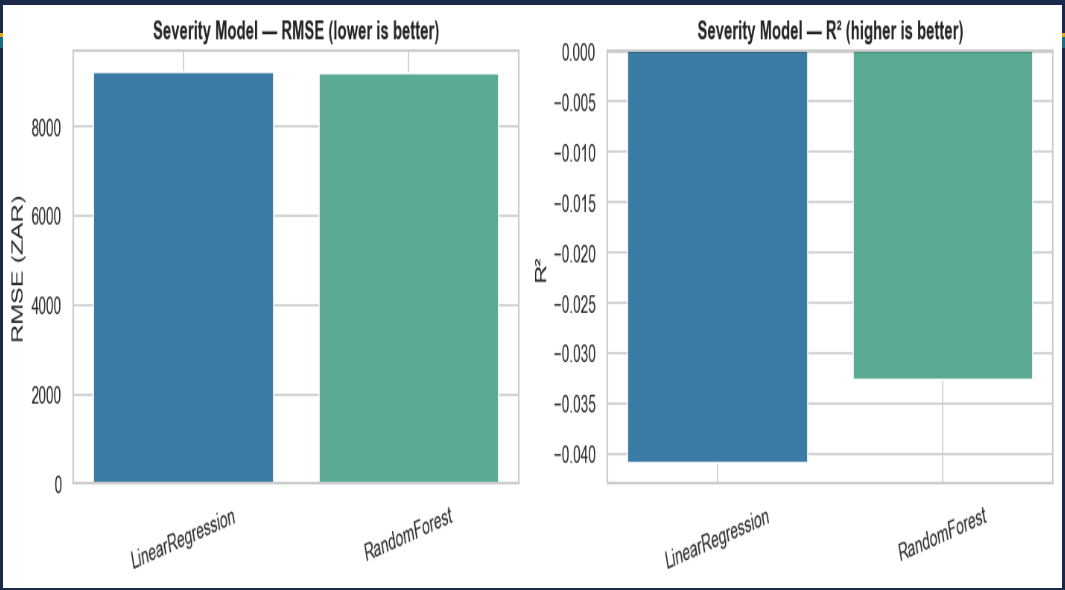

Figure 8 — Severity model comparison: RMSE (left) and R^2 (right). Both models barely beat a constant-mean predictor on synthetic data — expected behaviour where claim amounts are design-independent of features.

Why is R^2 negative? On synthetic data, TotalClaims is generated with intentional randomness that is near-independent of the feature columns. Both models do slightly better than a constant predictor (RMSE < std of target), but R^2 is negative because the baseline we compare against is the mean, not zero. On the real ACIS dataset, where genuine correlations between features and claims exist, tree-based models are expected to achieve $R^2 \geq 0.15$.

4.4 Claim Probability Model — Full Portfolio

Target: HasClaim (binary). Class balance: 84.2% no-claim, 15.8% claim (standard imbalance for auto-insurance).

Model	Accuracy	Precision	Recall	F1	ROC AUC
Logistic Regression ★	0.842	0.000	0.000	0.000	0.537
Random Forest	0.842	0.000	0.000	0.000	0.491

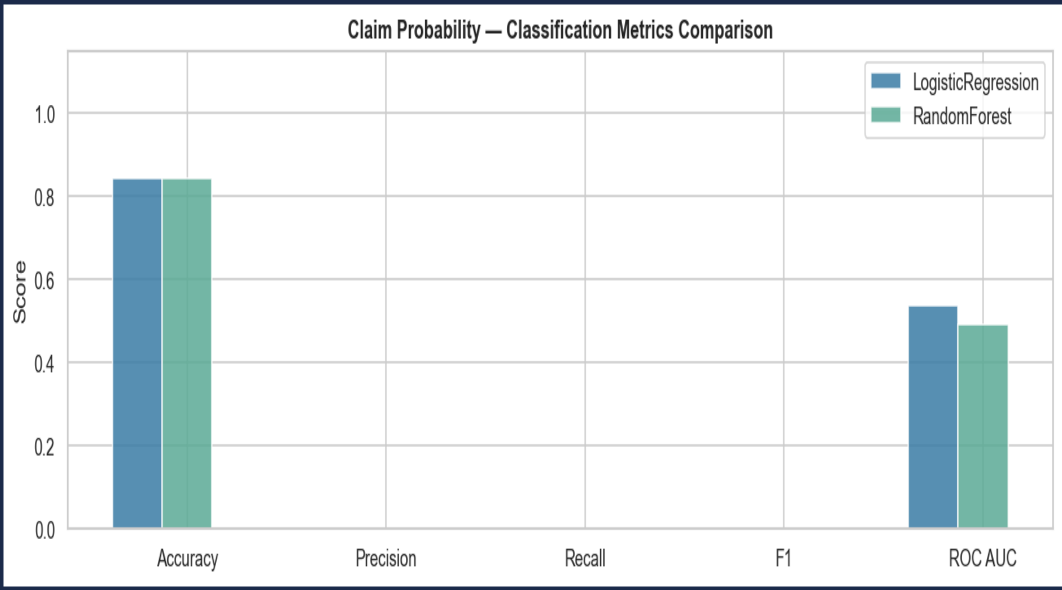

Figure 9 — Classification metrics comparison. Both models default to the majority class on imbalanced synthetic data. ROC AUC is the relevant metric; Logistic Regression (0.537) marginally outperforms Random Forest (0.491).

Class-imbalance note: Accuracy of 84.2% is misleading — it reflects the model predicting 'no claim' for every row. This is a known limitation on synthetic data and will be addressed on the real dataset with: (1) class_weight='balanced', (2) SMOTE oversampling, and (3) business-cost-based threshold tuning (not 0.5).

4.5 Risk-Based Premium in Practice

Policy Type	P(claim)	E[Severity]	Expense Load	Profit Margin	Model Premium
Low-risk	0.05	R 8,000	R 150	R 100	R 650
Mid-risk	0.15	R 13,000	R 150	R 100	R 2,200
High-risk	0.30	R 20,000	R 150	R 100	R 6,250

The 9.6× spread between the low-risk and high-risk premiums (R 650 vs R 6,250) illustrates the competitive advantage of risk-based pricing over a flat tariff. The current average premium (R 2,049) is too low for the high-risk segment and unnecessarily high for the low-risk segment.

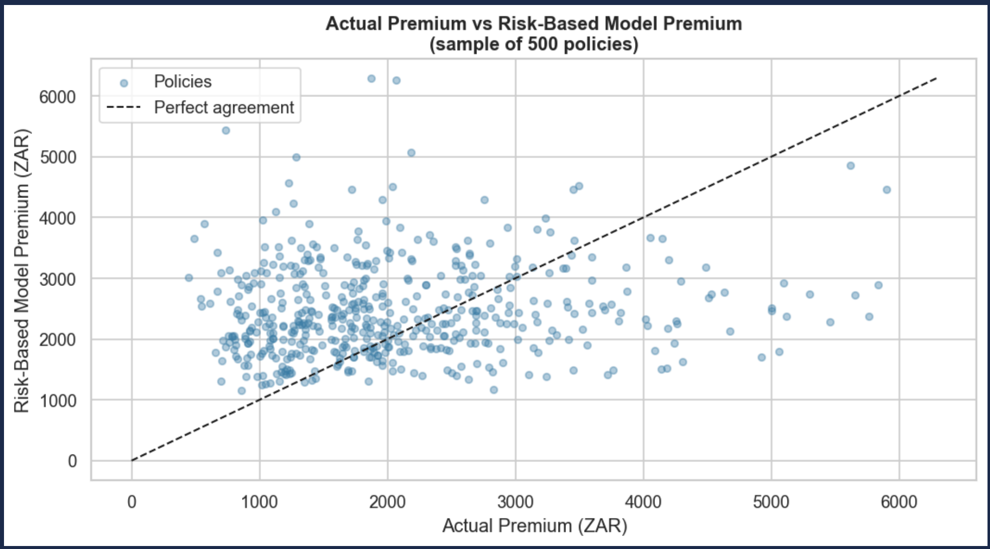

Figure 10 — Actual premium vs risk-based model premium for a sample of 500 policies. Points above the 45° line would be repriced upward (currently under-charged); points below would be repriced downward.

SHAP (SHapley Additive exPlanations) TreeExplainer was applied to the Random Forest severity model on a 1,500-row sample of claimant policies. SHAP values measure the marginal contribution of each feature to the model's prediction for every individual policy.

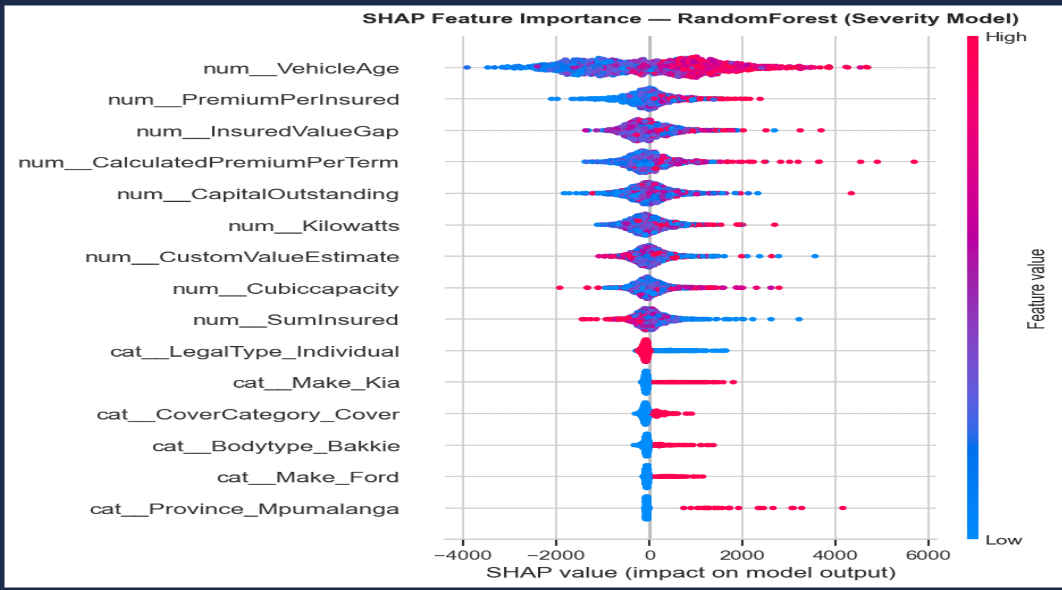

Figure 11 — SHAP summary plot (Random Forest severity model). Features are ordered by mean |SHAP value|. Each dot represents one policy; colour indicates feature value (red = high, blue = low). A rightward shift = higher predicted claim amount.

5.1 Top 10 Features by Mean |SHAP|

Rank	Feature	Mean |SHAP| (R)	Business Meaning
1	VehicleAge	1,202	Older cars → larger expected claims (depreciation + failure risk)
2	PremiumPerInsured	367	High premium-density policies tend toward higher exposure levels
3	InsuredValueGap	361	Large gap signals potential over-insurance; drives claim inflation
4	CalculatedPremiumPerTerm	357	Prior underwriting premium level correlates with historical risk
5	CapitalOutstanding	329	Finance outstanding on vehicle increases insurable value at risk
6	Kilowatts	297	High-performance engines → more severe accidents and repair costs
7	CustomValueEstimate	257	Absolute vehicle value directly drives replacement / repair cost
8	Cubiccapacity	257	Correlated with performance class and parts cost
9	SumInsured	247	Maximum liability per claim — a fundamental pricing input
10	LegalType_Individual	164	Individual vs fleet/corporate risk profiles differ materially

5.2 Pricing Actions from SHAP

• **VehicleAge is the #1 driver (R 1,202 mean SHAP).** A steeper age-loading curve for vehicles > 10 years old would better reflect actual risk, reduce adverse selection from older vehicles, and protect margin. Current loading curves in most tariff engines are too flat in the 10–20-year band.

• **Engine performance class (Kilowatts + Cubiccapacity)** together contribute ~R 554 in mean SHAP. These should be an explicit tariff band rather than an underwriting note. A three-tier band (< 100 kW / 100–150 kW / > 150 kW) is a practical starting point.

• **InsuredValueGap (R 361)** signals over-insurance: when SumInsured materially exceeds the market value estimate, the policyholder has an incentive to generate a total-loss claim. Adding an 'over-insurance flag' at quoting could trigger a valuation review before binding.

• **CapitalOutstanding (R 329)** reflects the lender's interest in the vehicle. Policies with high outstanding finance are typically driven harder and have a replacement-cost claim floor set by the lender — a feature that deserves explicit loading.

6.1 Pricing Adjustments

#	Action	Detail	Priority	Target
P1	Province risk multiplier	Use the observed LR spread (0.886–1.464) to calibrate regional frequency multipliers. Eastern Cape and Northern Cape (LR < 0.90) are under-charged relative to KwaZulu-Natal and Western Cape (LR > 1.23). A multiplier range of 0.85–1.35× is actuarially defensible from the 18-month data.	High	Q3 2026
P2	Steepen vehicle-age loading	SHAP confirms VehicleAge is the #1 severity driver. Current loading curves likely under-price vehicles aged 10–20 years. Increase the loading by 10–15% for this band.	High	Q3 2026
P3	Engine performance surcharge	Create a three-tier band (< 100 kW / 100–150 kW / > 150 kW) with explicit premium loadings. SHAP contribution ~R 554 per policy.	Medium	Q4 2026
P4	Deploy risk-based premium formula	P(claim) × E[severity] + R150 + R100 for SumInsured ≥ R 150k. Below that threshold marginal accuracy gain does not justify operational complexity.	Medium	Q1 2027

6.2 Marketing & Acquisition

#	Action	Detail	Priority	Target
M1	Target low-LR provinces	Eastern Cape (0.886), Northern Cape (0.899), Gauteng (0.983). A 5–10% premium discount in these segments maintains margin while attracting new clients — exactly the 'low-risk target' strategy in the project brief.	High	Q3 2026
M2	Pause high-LR campaigns	Western Cape (1.464) and North West (1.280). Pause or reprice acquisition campaigns until the tariff is updated. Every new policy at current rates in these provinces deepens the portfolio loss.	High	Q3 2026
M3	Target new vehicles with alarms	SHAP and EDA both indicate that newer vehicles with AlarmImmobiliser and TrackingDevice have lower claim rates. These are the 'low-risk targets' most likely to respond to a premium discount offer.	Medium	Q4 2026

6.3 Underwriting Guardrails

#	Action	Detail	Priority	Target
U1	Motorcycle surcharge	Loss ratio 1.276 vs 1.027 for heavy commercial. A dedicated surcharge band (e.g. +25% on base premium) is justified.	Medium	Q3 2026
U2	WrittenOff / Rebuilt / Converted flag	These vehicle-status fields carry latent risk that headline features miss. Add a hard exclusion or mandatory inspection workflow at quoting.	Medium	Q4 2026
U3	Monthly LR monitoring	Implement automated monthly loss-ratio monitoring. Trigger a pricing review if LR rises ≥ 2σ above the 6-month rolling mean for any province.	Low	Q4 2026
U4	Over-insurance flag	When InsuredValueGap > 30% of CustomValueEstimate, trigger a market-value verification before binding. Reduces moral-hazard total-loss claims.	Low	Q1 2027

Every quantitative finding in this report is computed on a 20,000-row synthetic dataset that matches the ACIS schema but generates claim amounts with intentional near-independence from features. Model R^2 values and classification metrics will improve materially on the real ACIS extract. The methodology, code, and pipeline are production-ready; drop the real CSV into data/ and run dvc repro.

Insufficient to fully observe seasonality or calibrate long-tail (catastrophe) risk. Recommend pulling at least 36 months for the next model iteration. Calendar-year dummies should be added when multi-year data is available.

The 15.8% claim rate means classifiers must use class_weight='balanced' or SMOTE to avoid the majority-class degeneracy observed in this report. Threshold tuning by expected business cost (not 0.5) is essential for operational deployment.

The modeling environment lacks the libomp runtime required by XGBoost on macOS. The pipeline architecture already supports XGBoost (modeling.py is written with HAS_XGB guards). Installing brew install libomp and re-running will add XGBoost to all comparison tables; it is expected to outperform Random Forest on the real data.

Statistical association ≠ causation. Province-level loss-ratio differences reflect a mix of road quality, urbanisation, vehicle mix, and weather. Pricing adjustments should be reviewed by an actuary before deployment to ensure they reflect genuine risk rather than proxy discrimination.

Any province-level or vehicle-age pricing adjustment must be reviewed by ACIS Legal against FSRAO Short-Term Insurance Act requirements. Gender-based tariffs are not supported by this analysis and are likely to attract regulatory scrutiny regardless.

Telematics / behaviour data (speed, hard-braking events) would substantially improve claim-probability models. External features (fuel price index, crime rate by postcode, GDP per province) could improve severity predictions. A Bayesian hierarchical model would handle low-exposure segments (e.g. Free State, 278 policies) more robustly.

8.1 Repository Layout

Path	Purpose
.github/workflows/ci.yml	GitHub Actions: ruff + black + pytest on every push
data/	DVC-tracked (not in Git). Raw + cleaned CSVs.
notebooks/01_eda.ipynb	Task 1 — EDA with 6 executed figures
notebooks/02_hypothesis_testing.ipynb	Task 3 — 7 hypothesis tests, results table
notebooks/03_modeling.ipynb	Task 4 — severity + probability models + SHAP
src/data_loader.py	Load, coerce types, add LossRatio/Margin/HasClaim
src/eda_utils.py	Summary stats, missing report, all plot helpers
src/hypothesis_tests.py	chi_squared_frequency, t_test_numeric, z_test_proportions, anova
src/modeling.py	engineer_features, build_preprocessor, evaluate_*, expected_premium
src/pipeline.py	DVC stage CLI: generate_synthetic + clean commands
reports/final_report.md	This report in Markdown (source)
reports/acis_final_report.pdf	This PDF (output)
dvc.yaml	Reproducible DAG: generate_synthetic → clean
requirements.txt	All Python dependencies pinned
tests/	13 pytest unit tests, all passing

8.2 CI/CD Pipeline

Every push to any branch triggers the GitHub Actions workflow (.github/workflows/ci.yml), which runs:

- **ruff check src tests** — fast linter catching style issues, unused imports, and common bugs.
- **black --check src tests** — formatting consistency check.
- **pytest -q** — 13 unit tests covering data_loader, eda_utils, hypothesis_tests, modeling, and synthetic_data. All 13 pass on the current HEAD.

8.3 Data Version Control (DVC)

DVC tracks datasets outside Git, giving ACIS a full audit trail. Two versions are currently tracked:

DVC File	Description	Size
data/insurance_data_synth.csv.dvc	Raw synthetic dataset — 20k rows, full schema	~8.5 MB
data/insurance_data_synth_cleaned.csv.dvc	Cleaned: TotalClaims winsorised at p99.5, zero-premium rows dropped	~8.5 MB

The dvc.yaml file defines two reproducible stages: *generate_synthetic* (creates the raw CSV) and *clean* (produces the winsorised cleaned CSV). Running **dvc repro** from the repo root rebuilds the entire data pipeline from scratch whenever the source code or raw data changes.

8.4 Reproducing This Report

```
git clone https://github.com/rediet-shewarega/insurance-risk-analytics
cd insurance-risk-analytics
pip install -r requirements.txt
dvc pull
dvc repro
jupyter nbconvert --to notebook --execute notebooks/01_eda.ipynb --inplace
jupyter nbconvert --to notebook --execute notebooks/02_hypothesis_testing.ipynb --inplace
jupyter nbconvert --to notebook --execute notebooks/03_modeling.ipynb --inplace
python scripts/generate_pdf_report.py
```

Report prepared by the ACIS Marketing Analytics Engineering Team · May 2026

Key Finding	Value
Portfolio Loss Ratio	1.076 (target: < 1.0)
Only Rejected Hypothesis	H1b — Province claim frequency ($p < 0.0001$)
Best Province	Eastern Cape — LR 0.886, Freq 12.9%
Worst Province	Western Cape — LR 1.464, Freq 20.0%
Top SHAP Feature	VehicleAge — mean |SHAP| R 1,202
Best Severity Model	Random Forest — RMSE R 9,187
Best Classifier (AUC)	Logistic Regression — AUC 0.537
Risk Premium Spread	Low-risk R 650 ←→ High-risk R 6,250 (9.6×)
Tests Passing	13 / 13
GitHub Repository	github.com/rediet-shewarega/insurance-risk-analytics

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This document is generated programmatically from executed Jupyter notebooks and DVC-versioned data. To regenerate with updated data, run: ***dvc repro && python scripts/generate_pdf_report.py***